UL 96

STANDARD FOR SAFETY

Lightning Protection Components

UL Standard for Safety for Lightning Protection Components, UL 96

Fifth Edition, Dated May 12, 2005

Summary of Topics

Revision pages have been issued for the Standard for Lightning Protection Components, UL 96, to reflect the extension of the effective date for paragraph 6.8. No changes in requirements have been made.

Text that has been changed in any manner or impacted by UL's electronic publishing system is marked with a vertical line in the margin. Changes in requirements are marked with a vertical line in the margin and are followed by an effective date note indicating the date of publication or the date on which the changed requirement becomes effective.

The requirements in this Standard are now in effect, except for those paragraphs, sections, tables, figures, and/or other elements of the Standard having future effective dates as indicated in the note following the affected item. The prior text for requirements that have been revised and that have a future effective date are located after the Standard, and are preceded by a ″SUPERSEDED REQUIREMENTS″ notice.

The following table lists the future effecive dates with the corresponding reference.

Future Effective Dates	References
May 12, 2006	Paragraph 29.2
October 10, 2014	Paragraph 6.8

ANSI/UL 96-2010

UL 96

Standard for Lightning Protection Components

First Edition – February, 1977
Second Edition – May, 1981
Third Edition – September, 1985
Fourth Edition – May, 1994

Fifth Edition

May 12, 2005

This ANSI/UL Standard for Safety consists of the Fifth edition including revisions through September 3, 2013.

The most recent designation of ANSI/UL 96 as an American National Standard (ANSI) occurred on October 4, 2010. ANSI approval for a standard does not include the Cover Page, Transmittal Pages, Title Page, or effective date information.

Comments or proposals for revisions on any part of the Standard may be submitted to UL at any time. Proposals should be submitted via a Proposal Request in UL's On-Line Collaborative Standards Development System (CSDS) at http://csds.ul.com.

No Text on This Page

CONTENTS

INTRODUCTION

CONSTRUCTION

CLASS I COMPONENTS

CLASS II COMPONENTS

CLASS II MODIFIED COMPONENTS

PERFORMANCE

INSTRUCTIONS

MARKINGS

No Text on This Page

INTRODUCTION

1 Scope

1.1 These requirements cover lightning protection components for use in the installation of complete systems of lightning protection on buildings and structures.

1.2 These requirements do not cover the installation of lightning protection components. Products of this type are covered by the Standard for Installation Requirements for Lightning Protection Systems, UL 96A.

1.3 Lightning protection components are divided into three classes, according to their intended application as defined in 4.3 – 4.5.

2 Units of Measurement

2.1 Values stated without parentheses are the requirement. Values in parentheses are explanatory or approximate information.

3 Undated References

3.1 Any undated reference to a code or standard appearing in the requirements of this standard shall be interpreted as referring to the latest edition of that code or standard.

4 Glossary

4.1 For the purpose of this standard the following definitions apply.

4.2 AIR TERMINAL – A component of a lightning protection system that is intended to intercept lightning flashes.

4.3 CLASS I COMPONENTS – All conductors, fittings, and fixtures necessary to protect ordinary buildings and structures not more than 75 feet (23 m) high.

4.4 CLASS II COMPONENTS – All conductors, fittings, and fixtures necessary to protect ordinary buildings and structures more than 75 feet (23 m) high.

4.5 CLASS II MODIFIED COMPONENTS – Components constructed for use in a system to protect a heavy duty stack.

4.6 CONDUCTOR – The portion of a lightning protection system intended to carry the lightning discharge between air terminals and ground.

a) Main Conductor – A conductor that interconnects air terminals and serves as a down lead to ground.

b) Secondary Conductor – A conductor that connects metal bodies to the lightning protection system to eliminate the buildup of an electrical potential between them during a lightning strike.

4.7 HEAVY-DUTY STACK – A smoke or vent stack that is more than 75 feet (23 m) high and has a flue with a cross sectional area greater than 500 square inches (0.3 m^2).

CONSTRUCTION

CLASS I COMPONENTS

5 General

5.1 Class I components shall be made of copper, copper alloy, aluminum or aluminum alloy with hardware made from stainless steel, unless otherwise required in this Standard, as outlined below:

a) Copper conductors, air terminals and stampings shall be made from electrical grade copper, C11000, generally designated as being 95% conductivity when annealed.

b) Aluminum conductors shall be made of electrical grade aluminum, with a minimum chemical composition of 99% aluminum.

c) Aluminum air terminals, stampings and couplings, shall be made with an alloy having a minimum chemical composition of 90% aluminum.

d) Stainless Steel hardware, such as nuts, bolts, washers, screws, threaded rods, and fasteners shall be of 18-8 grade (Chromium & Nickel content) with acceptable alloys being 302, 303 and 304.

e) Copper Alloys suitable for use in castings shall have a minimum copper content of 80%.

f) Aluminum alloys suitable for use in castings shall have a minimum aluminum content of 85%.

g) Brass alloys suitable for use in couplings, connectors, bases and fittings shall have a minimum copper content of 60%.

5.1 revised October 4, 2010

6 Air Terminals

6.1 An air terminal may be of one piece construction or may have several separate parts including: point (tip portion), elevation conductor, and base support.

6.2 An air terminal shall not be less than 10 inches (254 mm) long and shall comply with the requirements in Table 6.1.

Table 6.1
Minimum air terminal dimensions

Material	Construction	Diameter[a] inch	Diameter[a] (mm)	Base end thread diameter inch	Base end thread diameter (mm)	Wall thickness inch	Wall thickness (mm)	Cross-Sectional area[a] $inch^2$	Cross-Sectional area[a] $(mm)^2$
Copper and copper alloy	Solid	3/8	(9.5)	3/8	(9.5)	–		0.110	(71)
	Tubular	5/8	(15.9)	1/2	(12.7)	0.032	(0.81)	–	
Aluminum	Solid	1/2	(12.7)	1/2	(12.7)	–		0.196	(126)
	Tubular	5/8	(15.9)	1/2	(12.7)	0.064	(1.63)	–	

[a] The minimum diameter and minimum cross-sectional area are to be determined by measurements taken at various points along the axis of the air terminal for a distance not to exceed 50 percent of the total length of the air terminal measured from the threaded or base end and exclusive of the threaded portion or adapter of a tubular air terminal.

6.3 Deleted October 4, 2010

6.4 Each air terminal shall be provided with an integral base support or with not less than five full threads for attachment to the base support.

6.5 The threaded portion of an internally threaded air terminal shall have a minimum wall thickness of 1/16 inch (1.6 mm), measured at the base of the threads.

6.6 A tubular air terminal shall be provided with a threaded adapter for attachment to the base support. The threaded adapter shall comply with the requirements in 6.4 and Table 6.1 with respect to minimum thread size, and shall be securely attached to the air terminal.

6.7 The wind-resistance area of an ornament or decoration on a freestanding, un-braced air terminal shall not exceed 20 square inches (130 cm^2) in any plane. For example: A ball 5 inches (127 mm) or less in diameter complies with this requirement.

6.8 Any decoration, ornament or accessory added to the top section of an air terminal, shall be a minimum of 3/16 inch (4.8mm) thick and comply with 6.7 for wind resistance.

Added 6.8 effective October 10, 2014

6.9 A spring loaded air terminal adapter shall meet the following criteria:

a) The materials used shall be stainless steel and comply with the Standard Specification for Stainless Steel Spring Wire ASTM A 313/A 313-95a.

b) The spring shall be a closed spring with a minimum wire OD of .120 inches (3.048 mm).

c) The materials used for the adapter sections of the component shall meet the criteria in Section 5.

d) The design of the adapter sections of the component that attach to the air terminal and base shall comply with 6.4 and 6.5.

e) The adapter section of the component shall have a minimum length of 1-1/2 inches (38.1 mm).

f) The adapter section of the component shall have a minimum diameter of 5/16 inches (7.94 mm) and a minimum surface contact with the interior or exterior of the spring, along the axis of the spring, for 1/2 inches (12.7 mm) for each end of the adapter.

g) Each spring loaded air terminal component shall have a breakaway wire securely attached to it.

6.9 added October 4, 2010

7 Air Terminal Base Supports

7.1 The thickness of an air terminal base support shall not be less than the applicable value specified in Table 7.1.

Table 7.1
Thickness of air terminal base supports

Material	Construction	Minimum thickness inch	(mm)
Copper or copper alloy	Cast	3/32	(2.4)
	Stamped	0.061	(1.55)
Aluminum	Cast	3/32	(2.4)
	Stamped	0.097	(2.46)

7.2 A threaded hub provided for the attachment of the air terminal shall have at least five full threads and, if internally threaded, shall have a wall thickness of not less than 1/16 inch (1.6 mm) measured at the base of the threads.

7.3 A threaded hub is not required on a base support for a construction in which the air terminal is secured by a locknut on each side of the base support.

7.4 Each base support shall incorporate a connector fitting for connection to the lightning conductor. The conducting cross-sectional area of the base support, between the connector and the base of the air terminal, shall be equal to or greater than that of the conductor. The conductor shall contact the base for not less than 1-1/2 inches (38 mm) on all sides of the cable.

Exception: This requirement does not apply to a base support that complies with the Standard for Qualifying Permanent Connections Used in Substation Grounding, IEEE Std. 837. The applicable tests of IEEE Std. 837 shall be conducted with commercially available lightning protection conductors.

7.5 At least two mounting holes that will accept a No. 10 – 24 or larger bolt or screw shall be provided in the support so that it can be permanently and rigidly fastened.

7.6 An adhesive base shall be constructed with a minimum footprint of 7 square inches (45.1612 cm^2) overall. These bases shall have a minimum of six openings of 1/4 inch (6.35 mm) or greater and a minimum width of 2 inches (50.8 mm).

7.6 added October 4, 2010

7.7 An air terminals swivel coupling shall meet the following criteria:

a) The materials used shall comply with 5.1 and constructed so as to comply with 6.4 and 6.5.

b) Shall have a minimum length of 1-1/2 inches (38.1 mm) per section or segment.

c) Shall have a minimum surface contact area of 3/8 inch (9.525 mm).

d) Shall have a minimum bolt thread size of 1/4-20 that is made of stainless steel.

7.7 added October 4, 2010

7.8 An air terminal coupling shall meet the following criteria:

a) The material used shall comply with 5.1 and constructed in compliance with 7.2.

b) It shall have a minimum length of 1-1/4 inches (31.5 mm).

7.8 added October 4, 2010

8 Braces

8.1 An air-terminal brace shall be provided with two, three, or four legs and with one or two air-terminal guides.

8.2 A brace shall be made of 1/4-inch (6.4-mm) minimum diameter rod of aluminum, copper/copper alloy, stainless steel or hot-dipped galvanized steel.

8.3 Each mounting foot shall be flattened and shall have at least two mounting holes that accepts a No. 10 – 24 or larger bolt or screw.

8.4 A brace that is made of steel shall be protected from corrosion by hot dipped galvanized coating.

9 Conductors

9.1 Among the various types of lightning conductors are as follows: rope lay, smooth twist, and loose-weave cable; flexible and solid-strip conductors; tubular; and round, rectangular, square, or star-shaped rod.

9.2 The twist or lay of wires in a cable is not specified but the cable shall be stranded tightly enough to form a symmetrical cable and to remain in a fixed position when installed.

9.3 The size and weight of Class I conductors shall be as specified in Tables 9.1 and 9.2.

Table 9.1
Minimum dimensions of Class I main conductors

Type of conductor	Material	
	Copper	Aluminum
Cable		
Strand Diameter	0.045 inch (1.14 mm)	0.064 inch (1.63 mm)
Weight	0.187 pound/foot (278 gram/meter)	0.095 pound/foot (141 gram/meter)
Area	57,4000 circular mills (29 mm^2)	98,600 circular mills (50 mm^2)
Solid Strip		
Thickness	0.051 inch (1.30 mm)	0.064 inch (1.63 mm)
Width[a]	1 inch (25.4 mm)	1.21 inch (25.4 mm)
Solid Rod		
Weight	0.187 pound/foot (278 gram/meter)	0.095 pound/foot (141 gram/meter)

[a] This is the minimum width for a strip without perforations. If perforated, the minimum intended width is to be increased by the diameter of the perforations.

Table 9.2
Minimum dimensions of secondary conductors

Type of conductor	Material	
	Copper	Aluminum
Cable		
Strand diameter	0.045 inch (1.15 mm)	0.064 inch (1.63 mm)
Number of strands	14	10
Solid Strip		
Thickness	0.051 inch (1.30 mm)	0.064 (1.63 mm)
Width[a]	1/2 inch (12.7 mm)	1/2 inch (12.7 mm)
Solid Rod		
Diameter	0.162 inch (4.11 mm)	0.204 inch (5.18 mm)

[a] This is the minimum width for a strip without perforations. If perforated, the minimum intended width is to be increased by the diameter of the perforations.

10 Connector Fittings

10.1 Connector fittings as specified in Sections 11 – 14 shall comply with the requirements in this Section.

10.2 A connector fitting shall be a casting or shall be stamped from sheet stock, and shall comply with the requirements in Table 10.1.

Exception: A connector fitting that complies with the Standard for Qualifying Permanent Connections Used in Substation Grounding, IEEE Std. 837, is not required to comply with these requirements. The applicable tests of IEEE Std. 837 shall be conducted with commercially available lightning protection conductors.

Table 10.1
Minimum dimensions of connector fittings

Material	Construction	Thickness		Number of full threads	Bolt size
		inch	(mm)		
Copper or copper alloy	Cast	3/32	(2.4)	4	1/4 inch-20
	Stamped	0.061	(1.55)	–	
Aluminum	Cast	3/32	(2.4)	4	1/4 inch-20
	Stamped	0.097	(2.46)	–	

10.3 A connector fitting shall be constructed so that a minimum of 1-1/2 inches (38 mm) of each conductor can be secured within the connector.

Exception: This requirement does not apply to connector fitting that complies with the Standard for Qualifying Permanent Connections Used in Substation Grounding, IEEE Std. 837. The applicable tests of IEEE Std. 837 shall be conducted with commercially available lightning protection conductors.

10.4 The fitting shall be provided with at least two 1/8 inch (3.2 mm) high projections on an interior surface that embed in the conductor when the connector is compressed around the conductor.

Exception: This requirement does not apply to connector fitting that complies with the Standard for Qualifying Permanent Connections Used in Substation Grounding, IEEE Std. 837. The applicable tests of IEEE Std. 837 shall be conducted with commercially available lightning protection conductors.

11 Bimetallic Connectors

11.1 A bimetallic connector shall be made of copper and aluminum or a copper alloy and aluminum or stainless steel. The joint between the metals shall be constructed to exclude moisture.

11.2 A barrier shall be provided within each connector to separate the two conductors. When a conductive barrier is used, the material shall have electrochemical potentials below the 0.6 V level as indicated on Table 11.1.

Exception: The use of stainless steel hardware as outlined in 5.1 is permissible for use in contact with aluminum components and conductors.

11.2 revised October 4, 2010

Table 11.1
Electrochemical potentials (V)

	Magnesium, magnesium alloys	Zinc, zinc alloys	80 tin/20 Zn on steel, Zn on iron or steel	Aluminium	Cd on steel	Al/Mg alloy	Mild steel	Duralumin	Lead	Cr on steel, soft solder	Cr on Ni on steel, tin on steel, 12 % Cr stainless steel	High Cr stainless steel	Copper, copper alloys	Silver solder, Austenitic stainless steel	Ni on steel	Silver	Rh on Ag on Cu, silver/gold alloy	Carbon	Gold, platinum
Magnesium, magnesium alloys	0	0.05	0.55	0.7	0.8	0.85	0.9	1.0	1.05	1.1	1.15	1.25	1.35	1.4	1.45	1.6	1.65	1.7	1.75
Zinc, zinc alloys		0	0.05	0.2	0.3	0.35	0.4	0.5	0.55	0.6	0.65	0.75	0.85	0.9	0.95	1.1	1.15	1.2	1.25
80 tin/20 Zn on steel, Zn on iron or steel			0	0.15	0.25	0.3	0.35	0.45	0.5	0.55	0.6	0.7	0.8	0.85	0.9	1.05	1.1	1.15	1.2
Aluminium				0	0.1	0.15	0.2	0.3	0.35	0.4	0.45	0.55	0.65	0.7	0.75	0.9	0.95	1.0	1.05
Cd on steel					0	0.05	0.1	0.2	0.25	0.3	0.35	0.45	0.55	0.6	0.65	0.8	0.85	0.9	0.95
Al/Mg alloy						0	0.05	0.15	0.2	0.25	0.3	0.4	0.5	0.55	0.6	0.75	0.8	0.85	0.9
Mild steel							0	0.1	0.15	0.2	0.25	0.35	0.45	0.5	0.55	0.7	0.75	0.8	0.85
Duralumin								0	0.05	0.1	0.15	0.25	0.35	0.4	0.45	0.6	0.65	0.7	0.75
Lead									0	0.05	0.1	0.2	0.3	0.35	0.4	0.55	0.6	0.66	0.7
Cr on steel, soft solder										0	0.05	0.15	0.25	0.3	0.35	0.5	0.55	0.6	0.65
Cr on Ni on steel, tin on steel, 12 % Cr stainless steel											0	0.1	0.2	0.25	0.3	0.45	0.5	0.55	0.6
High Cr stainless steel												0	0.1	0.15	0.2	0.35	0.4	0.45	0.5
Copper, copper alloys													0	0.05	0.1	0.25	0.3	0.35	0.4
Silver solder, Austenitic stainless steel														0	0.05	0.2	0.25	0.3	0.35
Ni on steel															0	0.15	0.2	0.25	0.3
Silver																0	0.05	0.1	0.15
Rh on Ag on Cu, silver/gold alloy																	0	0.05	0.1
Carbon																		0	0.05
Gold, platinum																			0

Ag Silver
Al Aluminium
Cr Chromium
Cd Cadmium
Cu Copper
Mg Magnesium
Ni Nickel
Rh Rhodium
Zn Zinc

NOTE – Corrosion due to electrochemical action between dissimilar metals which are in contact is minimized if the combined electrochemical potential is below about 0.6 V. In the table the combined electrochemical potentials are listed for a number of pairs of metals in common use; combinations above the dividing line should be avoided.

S3426D

12 Water-Pipe Connectors

12.1 A water-pipe connector shall be constructed so that it makes contact with the water pipe for a minimum contact length of 1-1/2 inches (38.1 mm) measured along the axis of the water pipe. The cable connection shall comply with 10.1.

Exception: This requirement does not apply to a waterpipe connector that complies with the Standard for Qualifying Permanent Connections Used in Substation Grounding, IEEE Std. 837. The applicable tests of IEEE Std. 837 shall be conducted with commercially available lightning protection conductors.

13 Ground-Rod Clamps

13.1 A ground-rod clamp shall not be less than 3/32 inch (2.4 mm) thick and shall be made of copper or copper alloy.

Exception: This requirement does not apply to ground-rod clamps that comply with the Standard for Qualifying Permanent Connections Used in Substation Grounding, IEEE Std. 837. The applicable tests of IEEE Std. 837 shall be conducted with commercially available lightning protection conductors.

13.2 The clamp shall contact the ground rod for a minimum contact length of 1-1/2 inches (38.1 mm) measured parallel to the axis of the ground rod. A clamp shall be secured with at least two bolts or cap screws.

Exception: This requirement does not apply to ground-rod clamps that comply with the Standard for Qualifying Permanent Connections Used in Substation Grounding, IEEE Std. 837. The applicable tests of IEEE Std. 837 shall be conducted with commercially available lightning protection conductors.

14 Bonding Plates

14.1 A bonding plate shall have a minimum thickness of not less than 3/32 inch (2.4 mm). The thickness shall not be less than 5/16 inch (7.9 mm) for bosses for screw threads.

14.2 A Class I bonding plate shall have a minimum surface contact area of 3 square inches (19.4 cm^2).

14.3 Secondary Class 1 and 2 bonding connectors shall have a minimum surface contact area of 0.021 square inches (0.133 cm^2) for Copper, and 0.032 square inches (0.208 cm^2) for Aluminum.

14.4 A cable connector fitting shall be provided as a part of each bonding plate.

14.5 Structural steel bonding plates shall have a thickness of not less than 3/32 inch (2.4 mm). The thickness shall not be less than 5/16 inch (7.9 mm) for bosses for screw threads, and shall have a minimum surface contact area of 8 square inches (51.6 cm^2).

15 Cable Mounting Clips and Fasteners

15.1 Clips for securing copper conductors shall be cast or made from sheet copper with a minimum thickness of 0.032 inch (0.81 mm) and a minimum width of 3/8 inch (9.5 mm).

15.2 Aluminum clips for securing aluminum conductors shall be of cast aluminum or made from sheet aluminum with a minimum thickness of 0.051 inch (1.3 mm) thick and a minimum thickness of 1/2 inch (12.7 mm) wide.

15.3 A pinch-type cable support shall be of substantial construction that can be closed by bending. If the support is provided with a screw shank, it shall be the equivalent of a No. 10 wood screw 1-1/2 inches (38 mm) long.

15.4 A fastener for securing a conductor to a masonry surface shall be of the expansion, drive-in, or fan-shank type; and shall be made of a material that is as resistant to corrosion as copper.

15.5 A tile surface fastener shall consist of a two-piece metal band provided with at least one 1/4 inch (6.4 mm) diameter bolt for securing the band to the tile.

15.6 The metal bands of a tile surface fastener shall have a minimum thickness of 0.051 inch (1.3 mm), a minimum width of 1 inch (25.4 mm), and shall be made of a material that is as resistant to corrosion as copper.

15.7 A slate fastener shall comply with the dimensional and material requirements in 15.6.

15.8 An adhesive fastener shall comply with 15.1, 15.2 and 15.3 and be constructed with a minimum footprint of 3 square inches (19.4 cm^2) overall. These fasteners shall have a minimum of four openings and a minimum width of 1-1/2 inches (38.1 mm).

15.8 added October 4, 2010

15.9 Plastic materials used in cable mounting clips and fasteners shall comply with the glue down hardware requirements for the adhesive, be UV resistant and suitable for a operating temperature range of -40 to 65.6 °C (-40 to 150 °F).

15.9 added October 4, 2010

16 Ground Electrodes

16.1 A copper-plate electrode shall have a minimum thickness of 0.032 inch (0.81 mm).

16.2 A plate electrode of iron or steel shall have a minimum thickness of 1/4 inch (6.4 mm).

16.3 A plate electrode shall have a minimum surface contact area of 2 square feet (0.186 m^2).

16.4 Rod and Pipe type electrodes are covered in the Standard for Grounding and Bonding Equipment, UL 467.

16A Thru-Structure Assemblies

16A added October 4, 2010

16A.1 A Thru-Structure Assembly shall meet the following criteria:

a) The minimum diameter of rod used in the assembly shall comply with Table 6.1. If stainless steel is utilized, the minimum diameter shall be 1/2 inches (12.7 mm).

b) Any PVC used in these assemblies shall be UV resistant, electrical conduit and shall be constructed in accordance with the Standard for Schedule 40 & 80 Rigid PVC Conduit and Fittings, UL 651.

c) Any EPDM, plastic or other non-metallic components used in these assemblies shall be UV resistant.

d) All other components used in these assemblies shall be comprised of air terminals, bases, connectors, couplings and other hardware that shall comply with the appropriate sections of this standard.

No Text on This Page

CLASS II COMPONENTS

17 General

17.1 Class II components shall comply with the requirements in 5.1, 6.3 – 6.5, 6.7, 7.1, 7.2, 7.4, 7.5, 9.2 and Sections 10 – 13, 15, 16, and 18 – 20.

17.2 Class II connector fittings shall be provided with threaded bolt that secures the cable as specified in Section 10, Connector Fittings.

18 Air Terminals

18.1 A Class II air terminal shall be made of copper, copper alloy, or aluminum, and shall be of solid construction.

18.2 The minimum diameter of a Class II air terminal shall be 1/2 inch (12.7 mm) if made of copper and 5/8 inch (15.9 mm) if made of aluminum. The minimum diameter shall be determined as described in note a to Table 6.1.

18.3 An air terminal shall be at least 10 inches (254 mm) long.

18.4 An air terminal shall be provided with at least five full threads for attachment to the base support. The major diameter of the thread shall be equal to or greater than the minimum diameter of the air terminal.

19 Conductors

19.1 A Class II conductor shall be a cable made of copper or aluminum and shall comply with the requirements in Table 19.1.

Table 19.1
Minimum dimensions for Class II conductors

Type of Conductor	Material	
	Copper	Aluminum
CABLE		
Strand Diameter	0.0571 inch (1.45 mm)	0.072 inch (1.83 mm)
Weight	0.375 (pound/foot) (558 grams/meter)	0.190 (pound/foot) (283 grams/meter)
Area	115,000 Circular mills (58 mm^2)	192,000 Circular mills (97 mm^2)
SOLID STRIP		
Thickness	0.064 inch (1.63 mm)	0.1026 inch (2.61 mm)
Width[a]	1.40 inch (35.58 mm)	1.462 inch (37.16 mm)

[a] This is the minimum width for a strip without perforations. If perforated, the minimum intended width is to be increased by the diameter of the perforations.

20 Bonding Plates

20.1 A Class II bonding plate shall comply with the material and dimensional requirements for Class I bonding plates.

20.2 A bonding plate for utilizing the steel framework as a conductor shall have a surface contact area of not less than 8 square inches (52 cm^2).

20.3 A connector fitting shall be provided as a part of each bonding plate. See Connector Fittings, Section 10.

CLASS II MODIFIED COMPONENTS

21 General

21.1 Class II modified components shall comply with the requirements in 6.4, 6.5, 6.7, 7.2, 7.4,7.5, 10.2 – 10.4, 15.4, 21.2, and Sections 12 – 14, 16, 20, and 22 – 26.

21.2 All Class II modified components except the lightning conductor shall be made of copper, copper alloy, stainless steel, or other metal having equivalent resistance to corrosion as stainless steel.

22 Air Terminals

22.1 A Class II modified air terminal shall be made of solid copper, copper alloy, stainless steel, or monel metal having a diameter of not less than 5/8 inch (16 mm) and shall not be less than 18 inches (457 mm) long.

22.2 An air terminal made of copper or copper alloy shall be protected from corrosion with lead coating or equivalent.

22.3 Each air terminal shall be provided with five full threads for attachment to a base support. The thread diameter shall not be less than 1/2 inch (12.7 mm).

23 Air-Terminal Supports

23.1 A Class II modified air-terminal support shall have a thickness of not less than 1/8 inch (3.2 mm). The thickness shall not be less than 5/16 inch (7.9 mm) for bosses for screw threads.

24 Conductors

24.1 A conductor shall be a copper cable weighing not less than 0.375 pounds per foot (560 g/m). No strand of the cable shall be smaller than 15 AWG (1.65 mm^2).

25 Connector Fittings

25.1 The thickness of each connector, splicer, and associated fitting shall not be less than 1/8 inch (3.2 mm). The thickness shall not be less than 5/16 inch (7.9 mm) for bosses for screw threads.

Exception: This requirement does not apply to connector, splicer, or associated fitting that complies with the Standard for Qualifying Permanent Connections Used in Substation Grounding, IEEE Std. 837. The applicable tests of IEEE Std. 837 shall be conducted with commercially available lightning protection conductors.

26 Bonding Plates

26.1 Each bonding plate shall be at least 3/16 inch (4.8 mm) thick and have a surface contact not less than 8 square inches (52 cm^2) for stacks having a structural steel framing or 3 square inches (19 cm^2) for other Class II modified structures.

26.2 Each bonding plate shall be provided with a connector fitting. See Connector Fittings, Section 10.

PERFORMANCE

27 Security of Components

27.1 A connector shall withstand a weight of 200 pounds (890 N) for 1 minute as described in 27.2.

27.2 A connector is assembled with the conductors as specified by the manufacture and rigidly supported. The assembly bolts shall be torqued to the manufacturer's specifications. A 200 lb weight is to be suspended from the conductor for 1 minute without damage to the connector or conductor. There shall be no movement of the conductor during this test. Each conductor is to be tested independently for multiple conductor connectors. The pull shall be exerted by means of a tension-testing machine, dead weights, or other equivalent means so that there is no sudden application of force or jerking during the test.

27.2 revised October 4, 2010

27.3 A ground-rod clamp shall withstand without damage, a tightening torque of 150 pound-inches (17 N·m) applied to each clamping bolt or screw. The tightening torque is to be applied to each clamping bolt or screw when assembled on each size of grounding electrode intended for use.

INSTRUCTIONS

28 Installation Instructions

28.1 Installation Instructions shall be provided for lightning protection components that require assembly.

MARKINGS

29 General

29.1 Lightning protection components shall be marked, where it will be plainly visible after installation, with the manufacturer’s name, trademark, or other descriptive marking by which the organization responsible for the product is identified.

Exception: Clips and fasteners may be provided with markings on the smallest unit packaging.

29.2 Bimetallic connectors shall be marked ″Bimetallic″, ″Bimetal″ or ″BM″ on the connector.

29.2 effective May 12, 2006

29.3 If a manufacturer produces or assembles lightning protection components at more than one factory, each piece shall have a distinctive marking by which it is identified as the product of a particular factory.